

Bhavya Pratap Singh Tomar

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EXPERIENCE

Software Engineer

Apr 2026 – Present

Nirvana Solutions Pvt. Ltd.

Remote

- Designed **Dynamic code analysis pipeline** for large-scale .NET codebase using **SCIP-generated call graphs** indexed into **Neo4j MCP** into **claude** skills with **BST traversal**; across **30 machines** through centralized pipeline automation.
- Developed **context-engineered analysis layer** via **Claude Code + Notebook LM MCP** with module/class **dependency graphs**, cutting **token cost by 46%** vs. naive grep via AI harness coding agents
- Built **root cause analysis with auto-debug pipeline** classifying bugs from automation test sheets, pinpointing exact files/locations, packaged as **/auto-debug claude skill**; architected a **PR reviewer agent** for the company AI harness.

AI Development Intern

Nov 2024 – Jan 2025

Seequenze Technologies

Noida, U.P.

- Designed **LLM-based code refactoring system** processing **200K+ tokens** via **AST-level chunking** to resolve context window limits; **FastAPI** backend sustained **10K+ concurrent users**.
- Shipped a novel **code modernization pipeline** converting static layouts to **responsive design** and inline CSS to **Tailwind CSS** via structured prompt chaining and engineering, adopted by **1,000+ business clients**.

Artificial Intelligence Intern

Sept 2024 – Nov 2024

FoodNEST(S)

Lucknow, U.P.

- Deployed **multilingual multimodal Agentic Waba bot** on **Watsonx Granite models** across **22+ Indic languages**, automating order workflows for **2M+ stakeholders** and cutting manual operational overhead by **30%**.
- Architected **RAG application** on **LlamaIndex + Watsonx.ai** with governance-compliant model monitoring for **500K+ clients**; built distributed **web/map/video scrapers** (Python, Docker, EC2) at **98% uptime**.

PROJECTS

CQ Lite | *LangGraph, FastAPI, Gemini, AST, Bandit, ChromaDB, Python*

Live Analyzer

- Orchestrated a **multi-agent workflow** using **LangGraph** for conditional routing between file discovery, static analysis (**AST, Bandit**), and **Gemini**-powered review agents.
- Deployed as a **FastAPI** service, enabling a **ChromaDB**-powered Q&A system with live repository/PR review analysis.

AlgoPharma | *FastAPI, Groq/Llama 3.3, Celery, Redis, HuggingFace, spaCy, medspaCy, Neo4j*

- Architected **agentic pharmacovigilance platform** with **MCP-driven dynamic query routing** (Groq Llama 3.3 70B) and **3-layer PII guard** (regex → clinical NER OpenMed 44M → verification) for HIPAA/GDPR-compliant processing.
- Implemented **4-gate ADE detection** (drug NER → disease NER → sentiment → **medspaCy negation**) with full audit logging while validating signals using **PRR and ROR** metrics; async **Celery/Redis** pipeline for horizontal scalability.

Fillado | *LangGraph, Neo4j, FastAPI, WebSockets, MCP, ArmorIQ*

Demo

- Engineered **market intelligence layer** ingesting vernacular news into **multi-agent LangGraph Agent Council**; synthesis agent writes validated causal links to **self-updating Neo4j Knowledge Graph**.
- Implemented **hallucination-detection middleware** monitoring agent outputs in real-time; exposed tools as **MCP server** with **ArmorIQ** intent-token validation and full audit trail on every tool call.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL, Solidity, Linux (Bash)

AI & Machine Learning: LLMs, RAG, Fine-tuning (SFT, GRPO), Prompt Engineering, Transformers, Deep Learning (CNN, RNN), NLP, GenAI (Granite Models), AgentOps, MCP

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, spaCy, LlamaIndex, Celery

Cloud & DevOps: IBM Watsonx.ai (Governance & Orchestration), Azure AI, AWS (EC2, S3, IAM), Docker, Redis

Databases: Vector (Pinecone, ChromaDB, Weaviate), NoSQL (MongoDB, Neo4j), SQL (SQLite, PostgreSQL)

Web Development: FastAPI, MERN (MongoDB, Express, React, Node.js), ThirdWeb, WebSockets, RESTful APIs

Specialized Areas: AI Governance, Ethical AI, MLOps (CI/CD), Model Monitoring, Clinical NLP, AST Parsing

Certifications: Coursera: ML Specialization (Stanford); NLP Specialization (DeepLearning.AI)

ACHIEVEMENTS

- Amazon - National Top 25** — Ranked **23/83K+** in ML Challenge for a **Two-Tower DL model** on product pricing 2025
- Intel - National Winner** — Top 12/4K+ in **AI Hackathon**; won **25k Rs** for building **Agentic AI using OPEA™** 2025
- Hackodisha 5.0 - National Winner** — Won **\$170** for Best AI Project among 6,000+ participants for **Neon-Canvanies** 2025
- AI for Bharat - National Top 2** — Among **13k** for **AlgoPharma** pharmacovigilance platform under CDSCO central body 2026
- Google Cloud Agentic AI Day - National Finalist** — Among 59000+ at BIEC for **Farmers Agentic Solution** 2025
- AMD - National Top 10** — Among 3K+ registrations in **AI Sprint** for **finetuning Llama & Qwen models** 2025
- BNB Chain - National Runner-up** — Out of 2K+ at IIT Delhi; won **9k Rs** for an **AI tool on blockchain** 2025
- Orange Vibe Jam - Bounty Winner** — Int'l 15th/2K+ registrants; won **\$121** for deploying a **blockchain game** 2025
- Mercer | Mettl - Top 50** — Among 3K+ in AI Arena 2.0 for a **GenAI personalized retail chatbot** 2023
- Wells Fargo - National Finalist** — Among 890+ at IIT Madras for **crack detection software** 2023

EDUCATION

Madhav Institute of Technology and Science

Gwalior, MP

Bachelor of Technology in Artificial Intelligence and Machine Learning

Nov 2022 – Jun 2026